

MANASI BHANGALE

Software Engineer — AI/ML & Backend Development

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Profile Summary

Computer Engineering graduate specializing in Artificial Intelligence, Machine Learning, and Generative AI, with experience building LLM applications, RAG pipelines, AI agents, and computer vision systems. Skilled in Python, LangChain, FAISS, Ollama, Transformers, FastAPI, and deep learning frameworks to develop scalable AI solutions using LLMs, vector databases, and multimodal models.

Work Experience

SiliconMount Tech Services

Dec 2025 - Present

Software Developer Intern — Remote

- Developed an AI-powered smart proctoring system using Python, FastAPI, OpenCV, MediaPipe, and YOLOv8 for real-time candidate monitoring, facial analysis, and automated suspicious activity detection across multiple violation scenarios.
- Built an AI-powered chatbot for myBOQ to enable natural language interaction with BOQ documents, automate query resolution, and provide contextual responses by integrating backend services with LLM-based workflows.
- Performed business analysis for warehouse and inventory management systems by studying client workflows, documenting functional requirements, and analyzing processes including product master, purchase orders, batch/lot tracking, expiry management, repacking, and warehouse operations.
- Collaborated with clients and development teams to translate business requirements into technical solutions, contributed to BOQ estimation software enhancements, backend feature development, documentation, and deployment-oriented software practices.

Projects

AcademiQ — AI Academic Document Intelligence System (RAG)

[GitHub](#)

- Built the backend Retrieval-Augmented Generation (RAG) pipeline — including text chunking, embedding generation, FAISS vector indexing, and local LLM integration — for a document-based academic Q&A system supporting PDF, DOCX, TXT, image, and URL inputs (5 formats).
- Designed an overlapping chunking strategy (400-character chunks, 50-character overlap) and generated semantic embeddings with the all-MiniLM-L6-v2 Sentence Transformer, validated for high retrieval accuracy across a 500+ page academic document test set.
- Integrated FAISS similarity search with locally hosted LLMs (Mistral, Phi-3 Mini) via Ollama to ground answers in retrieved document context, reducing hallucinations versus standalone LLM responses while keeping inference on-device for privacy.
- Built the pipeline to support downstream features layered on top of it — including document Q&A, quiz/question generation, and research paper analysis — served through a Streamlit interface.

Onyx — Offline AI Voice Assistant

[GitHub](#)

- Built a fully offline, voice-activated AI assistant integrating wake-word detection, speech-to-text, local LLM inference, text-to-speech, and a real-time HUD overlay to deliver an end-to-end conversational AI experience without cloud services.
- Engineered a modular AI pipeline using Ollama-hosted **Qwen3:14B** and **Qwen2.5-VL**, enabling natural language interaction, vision-based screen analysis, web search, GitHub search, Wikipedia retrieval, and Windows system automation through tool calling.
- Developed a robust voice interaction pipeline using a custom-trained **openWakeWord** model, **Faster-Whisper** for speech recognition, and **Coqui XTTS-v2** with automatic GPU/CPU/Piper fallback, supporting natural offline voice conversations and voice cloning.
- Implemented persistent conversational memory using **ChromaDB**, along with coding utilities, semantic search, and session-aware context retention, while maintaining complete on-device processing and user privacy.

Skills

- Programming Languages:** Python, Java, JavaScript, SQL
- AI/ML & Generative AI:** Machine Learning, Deep Learning, NLP, Generative AI, RAG, LangChain, Prompt Engineering, Hugging Face, Scikit-learn, TensorFlow, Keras, XGBoost
- Data Analytics & Visualization:** Data Analysis, Data Cleaning & Preprocessing, EDA, Statistical Analysis, Pandas, NumPy, Power BI, Tableau, Advanced Excel
- Backend Development & MLOps:** FastAPI, Flask, REST APIs, API Integration, SQLAlchemy, Docker, Docker Compose
- Databases & Vector Stores:** MySQL, MongoDB, SQLite, FAISS
- Tools & Platforms:** Git, GitHub, Postman, Ollama, Swagger
- Core Computer Science:** DSA, OOP, DBMS, Operating Systems, Computer Networks

Education

Dr. Babasaheb Ambedkar Technological University, Lonere

Bachelor of Technology — Computer Engineering | CGPA: 8.13 | 2022 – 2026

Professional Certifications

- Data Science, Machine Learning, Deep Learning & NLP — Udemy
- Generative AI with LangChain & Hugging Face — Udemy
- Artificial Intelligence — AcmeGrade (Mood Indigo, IIT Bombay)

Achievements & Activities

- Open Source Contributor — GirlScript Summer of Code (GSSoC) 2025
- Cleared Internal Evaluation Round — Smart India Hackathon (SIH) 2025
- Active LeetCode Participant — Contest Rating 1,749 (Top 10.45%, Global Rank 89,427); 200+ problems solved, 74.81% acceptance.
- Technical Coordinator, GDSC DBATU — organized workshops and peer learning initiatives.